

**Session 1 Environment & Agriculture: Tuesday, 8th of September, (12.45-2.15)- (1.5 h)**  
**National Research center**



**Prof. Dr. Nour Shafik El-Gendy** is a Professor in the field of Environmental Sciences, Sustainable Development, Clean Energy and Nano-Biotechnology. She is the chairman of the Board of Directors of Central Analytical Labs – Egyptian Petroleum Research Institute (EPRI). She is the Former Head Manager of Petroleum Biotechnology Lab., the Former Acting and Vice Head of Process Design & Development Department, (EPRI). El-Gendy is Advisor to the Chairman of the Board of Trustees of October University for Modern Sciences and Arts, MSA University. She is a visiting professor in the School of Chemical Engineering of Sathyabama Institute of Science and Technology, India and Trivandrum Engineering Science & Technology (TrEST) Research Park, India. She is the Former Advisor for the Egyptian Minister of Environment. She is member of Scientific Research, Technology, and Cyber-security Committee, National Council for Women (NCW) of Egypt. El-Gendy is a member of the Higher Scientific Committee of the Arab Union for Sustainable Development and Environment. She is also a member of the Arab Council for Creativity and Innovation. El-Gendy is expert in the assessment of environmental pollution and its remediation, wastewater treatment, biofuels, petroleum bioupgrading, green chemistry, nanobiotechnology, valorization of wastes, biocorrosion and its mitigation. She has published 12 chapters, 10 books and 142 research papers, and supervised 29 MSc and PhD theses. Dr. El-Gendy is also an editor and reviewer in 92 and 252 international journals, respectively. She participated also as PI, Co-PI or research member in national and international research projects concerning with; biovalorization of wastes, solid waste management, bioethanol from lignocellulosic wastes; biofuels and value-added products from algae, application of nanobiotechnology in upgrading of petroleum and bioremediation

of polluted environment. Dr. El-Gendy participated in 70 international workshops, 68 international training courses, 186 international conferences and 27 international seminars. Dr. El-Gendy established 4 ongoing research schools; Petroleum and Environmental Biotechnology, Petroleum and Environmental Nanobiotechnology, Biofuels, and Valorization of Solid Waste Biomass. Dr. El-Gendy is a member in many national and international committees, associations and organizations concerned with petroleum industry, sustainable development, environmental health and sciences. She is in international collaboration with different foreign international universities, research institutes, and centers. She is listed in the Stanford-Elsevier global list of top 2% scientists 2025. She was honored in many scientific forums. She was awarded from Egypt the Women Appreciation Award in the field of "Water, Energy and Environmental Sciences" - 2024. She was also awarded the Regional Award of the Arab Union for Sustainable Development and Environment in the field of "Environment and Sustainable Development" - 2023. She was awarded from India the Research Excellence Award, Science father Women Researcher Award 2022 for the contribution and honorable achievement in innovative research, and the International Scientist Award 2021 on Engineering, Science and Medicine. El-Gendy has also been awarded the Best Women Scientist Award 2021 from College of Agriculture and Food Sciences, Florida Agricultural & Mechanical University, USA. She has been honored by the Official Portal of Department of Environment - Ministry of Environment and Water - Malaysia and University of Putra Malaysia - 2021. Her biography is recorded in Who's Who in Science and Engineering.

### **Marine Macroalgae Recruitment for Achieving Food, Energy and Water Nexus**

Intensive fishing practices, climate change, nutrient enrichment, waste discharge, sedimentation, and other forms of anthropogenic pollution contribute to the excessive proliferation of marine macroalgae, eutrophication, coral bleaching and mortality, and reduced light penetration for photosynthesis. These impacts disrupt marine ecosystems, diminish biodiversity, damage coastal environments, and adversely affect food webs, tourism, economic stability, and progress toward the Sustainable Development Goals (SDGs).

To advance the three pillars of sustainability—environmental protection, social well-being, and economic development—while addressing the food–energy–water nexus, it is essential to valorize surplus aquatic biomass through integrated, zero-waste, circular economy approaches. Our findings demonstrate that converting this underutilized biomass into value-added products is both feasible and environmentally beneficial. Specifically, it can be applied in (1) wastewater treatment; (2) the production of biodegradable plastics, biocides, probiotics, biofertilizers, and green nanocatalysts for diverse applications; and (3) the generation of various types of biofuels.



**Dr. Shaden Diab**, research professor and trainer at several academic and training institutions in France and the Euro-Mediterranean region, PhD in Environmental Sciences. She is an international expert in climate change and sustainable development. A chemical engineer, she holds a PhD in Environmental Sciences from Pierre and Marie Curie University (Paris 6). She has over fifteen years of experience in higher education, scientific research, training, and international consulting, working as a research professor and trainer at several academic and training institutions in France and the Euro-Mediterranean region. Her work focuses on climate change, climate justice, corporate social responsibility, educational innovation, artificial intelligence for climate, and community resilience. She is also a Climate Peace Ambassador for the Institute for Economics & Peace (IEP), where she focuses on the relationship between climate change, peace, and human security. She has received several international awards and honors related to environmental innovation and climate action, including contributions and recognition related to COP21. She was also selected for the Overseas Elite program in China in recognition of her contributions to innovation and international cooperation. Dr. Diab is considered a leading figure in the field of climate change and sustainable development. Shaden Diab is a member of the Scientific Committee of the European Federation of European Universities (FEDE) and contributes to developing visions related to education, quality training, and future skills. She also works as an international trainer in environmental journalism and climate communication, having delivered training sessions and lectures to journalists and media organizations, including national radio stations, the French Ministry of Foreign Affairs, the Labomed Laboratory, and the Arab States Broadcasting Union (ASBU), on the role of media in understanding climate challenges and communicating environmental issues to the public.

She is the founder of the Green Education and #GreenRefugees for Education project, which links environmental education, climate action, and the inclusion of vulnerable groups. This project has been developed in France, Greece, and Turkey. Through her writings, publications, and articles, she contributes to the global debate on climate change, climate migration, environmental security, artificial intelligence, and the transfer of green technologies between the North and South. Dr. Shaden Diab combines scientific expertise, academic work, and communication skills. Media, and climate diplomacy, with the aim of raising awareness and building innovative solutions to address major environmental challenges.

In addition to her scientific and academic career, Dr. Shaden Diab is a writer and novelist who is interested in combining science, literature, and human thought.

She has published several novels and intellectual works, including novels that address the relationship between humanity and the world and its transformations, such as the novel "A Chance with," where human questions meet scientific and philosophical reflections. In her latest book, "The Box... Yahya Who Saw the World: The Wonder of Creation and the Experience of Creation," she seeks to build a bridge between science and literature by exploring the relationship between creativity and knowledge, and between science fiction and human experience, presenting a vision that combines the physics of writing, the chemistry of creativity, and the embodiment of the text. This path reflects the distinctiveness of her experience, which combines scientific research, literary writing, and intellectual communication, with the aim of rethinking the relationship between science, culture, and humanity.

### **Environmental Diplomacy and Women: Issues of Stability and Climate Peace in the Arab World and the EMENA Region**

The world is confronting an acceleration of environmental crises — biodiversity loss, rising global temperatures, and the multiplication of climate "hotspots" (Pfenning-Butterworth, A., Buckley, 2024). Populations are growing risks of vulnerability and forced environmental migration. According to the World Bank's Groundswell reports, up to 216 million people could be displaced within their own countries by 2050, with Sub-Saharan Africa and North Africa among the most exposed regions. In this context, environmental diplomacy is emerging as a central issue for regional stability and security (Clemet 2021). This paper examines the role of women in environmental and climate diplomacy, with a particular focus on the Arab world and the EMENA/Africa region, where women are among the populations most vulnerable to climate change impacts. These vulnerabilities are reflected in increasing water insecurity, climate-induced displacement, food insecurity, and socio-economic pressures, which often exacerbate existing gender inequalities (IPCC, 2022; UN Women, 2022). Climate change affects women and men differently due to unequal access to resources, decision-making processes, land ownership, and economic opportunities. In many climate-vulnerable regions, women are often responsible for household water and food security, making them particularly exposed to environmental degradation and resource scarcity (UN Women, 2022; FAO, 2023). 1 At least one-third of refugee households in the Arab States are female-headed households, reflecting the growing responsibility placed on women in contexts of forced displacement. Female-headed households are often disproportionately affected by climate-related shocks because they have fewer economic resources, weaker social protection mechanisms and limited access to adaptation capacities (UN Women, 2020; UNHCR, 2023). The contribution of women to diplomacy has historical roots. Long before women gained formal access to diplomatic institutions, they played important informal diplomatic roles as salonnières, court intermediaries, political advisers, and spouses of ambassadors, using social networks and cultural spaces to facilitate communication, information exchange, negotiation and mediation between political actors (Calvié, 2015; Hamilton & Langhorne, 2011). These informal diplomatic practices demonstrate that women have historically contributed to international relations through forms of influence that operated outside traditional state structures and official diplomatic channels. The intersection between climate change, displacement and gender inequality

highlights the importance of integrating women's perspectives into environmental diplomacy, climate negotiations and regional security strategies.



**Dr. Pulane Hannah Koosaletse-Mswela** is an environmental scientist and Senior Lecturer at the University of Botswana, with expertise in environmental quality, water and wastewater management, waste management, and resource recovery. She holds a PhD in Environmental Science and an MSc in Applied Microbiology, and has built a strong academic and professional career at the intersection of science, sustainability, and policy-relevant research in semi-arid environments.

Her research focuses on environmental quality assessment, ecological risk analysis, and sustainability-oriented evaluation of water resources, waste systems, and anthropogenic activities, with particular emphasis on small dams, surface water bodies, and community-linked ecosystems in arid and semi-arid regions of Southern Africa. Her work supports evidence-based decision-making for environmental protection, public health, and sustainable recreational and productive water use.

Dr. Koosaletse-Mswela has supervised and graduated numerous postgraduate students at MSc and PhD levels, and has published in peer-reviewed journals while actively participating in national and international scientific conferences. She is experienced in interdisciplinary research, project coordination, and applied environmental assessment that bridges science and practice.

Beyond academia, she has contributed to environmental governance and science-policy interfaces, including serving on the technical committee that advised Botswana's National Waste Management Policy. She is the Secretary General of the Botswana Academy of Science and the inaugural Vice Chair of the OWSD-Botswana Chapter, where she advocates for mentorship, inclusive excellence, and leadership development for women in science.

Her work aligns strongly with global efforts to empower science, innovation, and entrepreneurship in addressing environmental challenges and building resilient, sustainable futures for society.

### **Environmental Quality-Based Assessment of a Small Dam for Sustainable Recreational Use in Semi-arid Botswana**

Tourism is one of the fastest-growing sectors globally and the Botswana Tourism Organisation has come up with the Dams Tourism initiative to diversify and grow the tourism sector in Botswana. The absence of national recreational water quality standards complicates the assessment of these environments for safe tourism use, hence the need for baseline data to guide sustainable planning and policy development. This study, therefore, assessed the environmental quality of Mogobane Dam, to determine its suitability for recreational eco-tourism. Water, sediment, and fish samples were analyzed for physico-chemical parameters, heavy metals, and fecal coliforms, while qualitative data were gathered through semi-structured interviews and questionnaires administered to community leaders, farmers, and tourists. Secondary data sources, such as rainfall and temperature records, were obtained from the Department of Meteorological Services. The results indicated that all analyzed water samples contained detectable levels of lead, iron, and aluminium. The mean Pb concentration (4.17 mg/L) exceeded WHO, BOBS, and SANS permissible limit. Fecal coliform testing recorded values beyond the WHO and USEPA guidelines for safe recreational waters, deeming Mogobane Dam unsuitable for direct contact activities such as swimming or consumption. Fish samples (*Clarias gariepinus*) contained elevated heavy metal levels, particularly in gills and fins. Interviews revealed limited awareness of water quality issues and the absence of formal management strategies among local users. No structured environmental monitoring or waste management practices were found in place. The Geo-accumulation Index classified Mogobane sediments as moderately to highly polluted, while Enrichment Factor values suggest significant anthropogenic contribution to heavy metal accumulation. Bioaccumulation Factor analysis indicated strong uptake of Pb and Fe in fish tissues, highlighting potential risks of metal transfer through the aquatic food chain. The study emphasises the necessity of regular water quality monitoring, watershed management and stakeholder awareness raising to facilitate the safe and sustainable utilisation of dams in Botswana and similar environments.





**Prof. Dr. Seham Ali Shaban** is a professor at the Egyptian Petroleum Research Institute. She has held significant leadership roles, including serving as the former head of the Refining Department, and the former head of the Catalyst Laboratory. Professor Shaban has supervised approximately 25 scientific theses. She has received numerous research grants and has published around 60 research papers in various fields, with a focus on green energy and waste treatment. In addition, she is a member of numerous international arbitration panels for scientific research and numerous local and international associations.

### **Biochar – Sustainable Material for A Green Future**

Biochar has created a lot of interest because of its unique properties of sustainable agricultural production and environmental protection. Biochar is known to reduce nitrogen loss from soil in terms of nitrous oxide emission and ammonia volatilization, improve the nutrient retention capacity and structural and chemical properties of soil, and increase plant growth and productivity. Biowaste materials such as biosolids, municipal waste, and paper mill sludge are effective raw materials for biochar production. These materials are rich in nutrient content, and biochar produced using these waste products is highly effective for agricultural production and environmental remediation.



**Dr. Safaa Saeid Mohamed Ali**, Associated Professor of Spectroscopy and its application. I had experience in Spectroscopic plasma diagnostics, Qualitative/quantitative determination of the concentration of elements in different kinds of samples, Trace and ultra trace analysis of liquid samples using the inductively coupled plasma (ICP-AES), Analysis of gaseous samples (hydride generation) using wall stabilized plasma arc and Elemental Analysis using atomic absorption and graphite furnace.

I am a staff member in number of research projects such as Egyptian-Spanish Joint Program of Co-operation on Industrial Research and Development “New Strategies for Reducing the Costs of the Steel Industry; Ladle Slag Recycling and Raw Material Refining” . As well as the Center of Training and Developing Skills, National Research Center, Cairo, Egypt, (Lecturer for Inductively coupled plasma and Atomic Absorption courses), also the Unit of Analysis and Central Scientific Service, National Research Center, Cairo, Egypt.

### **Spatial Distribution and Health Risk Assessment of Heavy Metals in the Road Dust Deposited on Tree Leaves in the main Overcrowded Squares of Greater Cairo**

Rapid urbanization and heavy vehicular traffic in Greater Cairo have led to increasing concerns regarding over heavy metal accumulation in urban dust and its potential impacts on public health. This study assessed the contamination levels and human risks of heavy metals in roadside dust deposited on tree leaves collected from highly crowded public squares across Greater Cairo, Egypt. Elemental concentrations were determined using inductively coupled plasma mass spectrometry (ICP-MS) and high-resolution continuum source graphite furnace atomic absorption spectrometry (HR-CS-GF-AAS). Contamination status was evaluated using enrichment factors (EF) and the Pollution Load Index (PLI). Non-carcinogenic and carcinogenic health risks for adults and children were estimated according to the US EPA risk assessment model.

Results showed that the average concentrations of all studied heavy metals exceeded local background soil values, indicating strong anthropogenic influence. PLI values greater than unity confirmed significant multi-element pollution across the sampled squares. Ingestion was the dominant exposure pathway, followed by dermal contact. Children exhibited higher non-carcinogenic risks than adult, particularly for cadmium. Although contamination levels were elevated, the carcinogenic risks values for Cr, Cd, and Pb remained within the acceptable limits ( $10^{-6}$  to  $10^{-4}$ ). These findings highlight the serious environmental burden of traffic-related heavy metals pollution in Greater Cairo's urban squares and underscore the urgent need for continuous monitoring and effective mitigation measures to protect public health in densely populated urban environments.



**Takya Mohammad Ahmed Elnady** is a Chemistry Teacher and Capstone Projects leader at El-Sadat STEM School, Egypt, and a PhD Researcher in the Department of Curriculum and Instruction at Sadat City University, where she combines classroom teaching with active research in science education. Her research interest in green chemistry education grew directly out of her classroom experience. While supervising students' capstone projects and practical laboratory sessions, she observed the significant volume of chemical waste generated by conventional experiments and the long time required for its decomposition. This observation shaped her research focus on integrating green, environmentally responsible alternatives into secondary science curricula. Building on this, she designed a structured program of green chemistry activities organized around five thematic axes, each



paired with a distinct active-learning strategy, aimed at developing two key learner outcomes: observable environmental practices and strategic thinking skills across five dimensions planning, generating creative alternatives, decision-making, systems analysis, and future foresight. At this forum, she will present her work title "The Role of Green Chemistry in Achieving Sustainable Development Goals: A Modern Educational Vision."

Beyond her doctoral work, Takya has also developed "EcoChem Lab," a poster presenting her green chemistry program for a school-level innovation competition at Cairo University, reflecting her continued effort to translate research into practical, classroom-ready tools. Takya's work sits at the intersection of chemistry education, environmental sustainability, and the Sustainable Development Goals, with particular focus on preparing STEM students to connect scientific knowledge with responsible environmental action.

### **The Role of Green Chemistry in Achieving Sustainable Development Goals: A Modern Educational Vision**

Amid escalating environmental challenges and the global drive toward achieving the Sustainable Development Goals, there is a growing need to reconsider science teaching curricula, moving beyond the transmission of theoretical knowledge toward building environmental practices and strategic thinking capable of addressing sustainability issues. The idea for this research emerged from field observations of the volume of chemical waste generated by students' experiments and practical projects, and the long time such waste takes to decompose — highlighting the need to move toward green alternative materials that achieve the same reactive purpose while reducing environmental impact. Within this framework, the researcher adopts an educational vision that employs the principles of green chemistry as an effective pedagogical entry point, rather than as isolated chemical content, aiming to link classroom learning with real-world environmental behavior among school students in general, and students of STEM (Science, Technology, Engineering, and Mathematics) schools of excellence in particular.

To address this problem, the researcher proposed a program based on green chemistry principles, from which a practical handbook to a number of alternative green chemistry experiments can be developed. This presentation discusses this program, which is designed around five thematic axes, with each activity linked to an active learning strategy, reflecting an intentional methodological diversity. The program aims to develop two variables among learners: environmental practices, understood as observable behaviors within the classroom and laboratory setting, and strategic thinking skills across their five dimensions (strategic planning, generating creative alternatives, strategic decision-making, systems thinking, and future foresight).

This modern educational vision positions green chemistry as a multidimensional pedagogical tool that contributes to achieving several Sustainable Development Goals — foremost among them Quality Education (Goal 4), Responsible Consumption and Production (Goal 12), and Climate Action (Goal 13) by preparing a generation of learners capable of connecting scientific knowledge with responsible environmental action.

**Session 2 Environment & Agriculture: Tuesday, 8<sup>th</sup> of September, 3.00-5.00 (2 h), National Research Center**



**Prof. Dr. Hamdy F. El-Mowafi** is an Emeritus Research Professor at the Field Crops Research Institute, Agricultural Research Center (ARC), Egypt, and the National Project Leader for the Development of Hybrid, Super, and Basmati Rice under Water Scarcity and Climate Change. He is one of Egypt's leading rice breeders, with over four decades of experience in rice genetics, breeding, and sustainable crop improvement.

Prof. El-Mowafi pioneered Egypt's hybrid rice breeding program and has developed numerous high-yielding, climate-resilient rice varieties, including hybrid, super, and Basmati rice cultivars. His research focuses on breeding for drought and salinity tolerance, water-use efficiency, and food security. He has authored more than 60 scientific publications and has received several prestigious awards, including the Gold Medal at the 48th Geneva International Exhibition of Inventions (2022) and the Arab Food Security Award (2023). Throughout his distinguished career, he has played a leading role in national and international rice research initiatives, collaborating with organizations such as the International Rice Research Institute (IRRI) and FAO, and has trained researchers and agricultural professionals across Africa and Asia.

**Climate-Resilient Super, Hybrid, and Basmati Rice Varieties for Sustainable Production under Water Scarcity and Climate Change**

Climate change, water scarcity, and increasing food demand pose significant challenges to sustainable rice production worldwide. Egypt has made remarkable progress in developing climate-resilient super, hybrid, and Basmati rice varieties through integrated conventional and modern breeding approaches. These varieties exhibit high yield potential, early maturity, tolerance to salinity, drought, heat, and limited irrigation, while maintaining superior grain quality. The developed cultivars, including Sakha Super 300, Sakha Super 301, Sakha Super 302, Sakha Super 303, Giza Basmati 201, and Egyptian Basmati Hybrid Rice 11, achieve yield advantages of up to 30% over traditional varieties and can save approximately

30–40% of irrigation water through improved water management practices. The use of EGMS and CMS breeding systems has further enhanced hybrid seed production and genetic purity. This chapter highlights Egypt's achievements in developing and disseminating climate-adapted rice cultivars and discusses their contribution to food security, sustainable agriculture, efficient water use, and climate change adaptation. These innovations provide a practical model for increasing rice productivity while strengthening the resilience of agricultural systems under changing environmental conditions.



**Jenny Carolina** lives in Cali, state of Valle del Cauca, in the country Colombia, located in South America. She is an Agricultural Engineer, holds a Master's degree in Biological Sciences and is PhD in Agroecology; currently is undertaking postdoctoral research within a program with the Colombian government that supports research projects by women in artificial intelligence, science and quantum technologies and is head of her house as single mom. A scientist with experience in public-private partnerships in agriculture and food systems programs. She has worked on research projects with multiethnic rural communities in Colombia and Spain. She has also contributed to the design, evaluation, and implementation of productive projects. In addition, she teaches at universities both nationally and internationally. She has pursued her postgraduate studies with scholarships and grants due to her high performance in research work. As part of the Agroecology Research Group at the National University of Colombia - Palmira campus, she has represented them in spaces such as the Academic Dialogue Platform on food and nutritional security as part of the Territorial Council for Food Sovereignty and Security for Santiago de Cali; she was a member of the Food and Nutritional Security Advisory Council, collaborating in the construction of the Food and Nutritional Security Action Plan of Valle del Cauca; she participated in micro-region workshops for the participatory construction of the "Departmental Plan of Agroecology of

Valle del Cauca" and in the Agroclimatic Technical Table of the agricultural sector of Valle del Cauca.

### **Smart Vermicomposting: Automating Biofertilizer Production Through Community Knowledge and Low-Cost IoT**

Synthetic chemical fertilizers have caused various environmental impacts, which is why there is increasing interest in the use of biofertilizers. However, the production of these fertilizers requires the incorporation of strategies that contribute to stabilizing their production. Therefore, this research project was proposed, consisting of two vermicomposting pilot projects: one at the biofertilizer plant of the Universidad Autónoma de Occidente (UAO) and the other on the ancestral territory of an Afro-Colombian community. The project focuses on automation, which allows for the monitoring and regulation of variables determining the quality of biofertilizer production within the process. The project's approach to biofertilizer production involves biotic component, utilizing red wiggler worms (*Eisenia foetida*) in vermicomposting processes to obtain solid and liquid humus. The automated system uses low-cost IoT devices to monitor crop variables (bed) such as temperature, humidity, and pH, and to generate regulatory actions. It also incorporates AI algorithms that process the variable values to generate estimates related to production levels. The social appropriation of IoT in the control of biofertilizer production variables is an ongoing process that requires the collaboration of multiple stakeholders and is a community-centered approach. Through the co-construction of knowledge mediated by participatory methodologies, significant benefits are generated for the Afro-Colombian community belonging to the "Sinécio Mina" Historical and Cultural Research Movement, associated with the project, as well as for the GEAS Research Group and the UAO's food security research group, by fostering an inclusive and creative use of these technologies, improving quality of life, and promoting sustainable development.



**Marwa Sulaiman Mohsin Al-Hinai** is a PhD researcher in Plant Sciences at Sultan Qaboos University, Oman, specializing in plant biotechnology, physiology, and abiotic stress tolerance. Her research focuses on the application of nanotechnology to enhance crop performance under drought and salinity stress, with particular emphasis on nano-trehalose formulations.

She holds a Master's degree in Plant Production (Plant Sciences) from Sultan Qaboos University, where her research centered on the characterization and improvement of wheat genotypes of Omani origin for salt tolerance. She also earned a Bachelor's degree in Applied Biology from the Higher College of Technology, Muscat. Her academic training has provided her with hands-on experience in plant stress physiology, antioxidant enzyme analysis, molecular techniques, tissue culture, and nanoparticle synthesis and characterization.

#### **Nanotechnology-Enabled Smart Agriculture for Enhanced Drought Resilience and Sustainable Agroecosystems**

Climate-induced drought has emerged as one of the most severe threats to global food security, disrupting soil–water dynamics and limiting crop productivity in arid and semi-arid regions. In this context, the convergence of nanotechnology and smart agriculture offers an unprecedented opportunity to develop climate-resilient and resource-efficient food systems. This review critically synthesizes recent advances in nano-enabled drought mitigation—focusing on nanosensors, nano-fertilizers, and nano-hydrogels—alongside digital technologies such as the Internet of Things (IoT), Artificial Intelligence (AI), and autonomous systems for precision water and nutrient management. The analysis reveals that integrating nanotechnology with smart farming frameworks enhances drought tolerance through molecular-level monitoring, controlled nutrient delivery, and data-driven irrigation optimization. Despite remarkable progress, significant challenges persist regarding nanoparticle safety, data interoperability, and scalability in resource-limited contexts. A conceptual Nano–Smart integration framework and a research roadmap are proposed to bridge current gaps and guide future innovation toward eco-safe, self-regulating, and ethically governed agroecosystems. This synthesis underscores that Nano–Smart agriculture is not merely a technological evolution but a transformative paradigm—linking scientific intelligence with ecological sustainability and social equity to secure food production under a changing climate.



**Dr. Ahmed R. Henawy** , Assistant Professor ,Cairo University & Huazhong Agriculture University. He holds a BSc in Biotechnology from Cairo University 2012, followed by an MSc

in Microbiology from Huazhong Agricultural University in China 2019, and a PhD in Microbiology from Cairo University 2023. He has also completed a MicroMasters in Bioinformatics through the University System of Maryland 2020, along with specialist training in metagenomics and whole-genome sequencing of bacteria. Currently, he serves as Assistant Professor in the Microbiology Department at Cairo University, a position he has held since 2023. Concurrently, he holds the role of Postdoctoral Fellow and Group Leader at Huazhong Agricultural University in Wuhan, China, where his research career first took an international dimension. His career reflects a strong commitment to bridging research and academia, to leading collaborative research groups at one of China's premier agricultural universities. He also serves as coordinator for academic collaboration between Cairo University and leading Chinese institutions, including HUST and HZAU.

**Gut Microbiomes of Black Soldier Fly Larvae: Approaches for Bioconversion, Degradation of Antibiotics, Detoxification of Mycotoxins, and Production of Biofertilizers for Sustainable Waste Management**

This study presents an innovative biotechnology framework that integrates Black Soldier Fly (*Hermetia illucens*) larval bioconversion with targeted microbial biodegradation to address critical challenges in organic waste management, food safety, and the development of a sustainable bioeconomy. The approach leverages the synergistic interaction between insect metabolism and specialised microbial consortia to transform agro-industrial residues into high-value products, including protein-rich biomass and biofertilisers.

A key focus of this research is the detoxification of mycotoxins contaminated substrates, wherein microbial strains and larval gut microbiota collaboratively facilitate degradation through enzymatic pathways such as oxidation, reduction, and hydroxylation. By combining culture-dependent microbial isolation with meta genomic and functional analyses, this research identifies key microbial players and metabolic pathways responsible for enhanced bioconversion efficiency and toxin mitigation.

The results demonstrate that integrating engineered or selected microbial communities with Black Soldier Fly systems significantly improves substrate conversion rates, detoxification efficiency, and system resilience under variable environmental conditions. These findings underscore the potential for scalable, cost-effective biotechnological solutions that can be translated into industrial applications.

Importantly, this research supports the development of circular economy models by converting waste streams into commercially valuable outputs, thereby opening opportunities for startup innovation and sustainable agribusiness. The study contributes to bridging the gap between laboratory research and market-oriented solutions, positioning insect-microbe systems as a promising platform for future bio-based entrepreneurship and green technology development.





**Dr. Asmaa Ahmed Mostafa Halema** is an Assistant Professor in the Genetics Department, Faculty of Agriculture, Cairo University, Egypt, a position she has held since June 2023. She earned her Ph.D. in Genetics from Cairo University (2019–2023) and her M.Sc. in Microbiology from Huazhong Agricultural University, China (2016–2019), under a Chinese Government Scholarship (CSC). She also holds a B.Sc. in Biotechnology from Cairo University, where she graduated first in her class.

Dr. Halema's research centers on microbial genetics, genomics, and bioinformatics, with a particular focus on heavy metal bioremediation, plant-microbe interactions, and omics-driven approaches to understanding bacterial resistance mechanisms. Her work spans whole-genome sequencing and analysis of environmentally significant bacterial strains, including *Enterobacter*, *Raoultella*, and plant growth-promoting rhizobacteria, contributing to sustainable agricultural and environmental biotechnology.

She has authored or co-authored numerous peer-reviewed publications in journals such as *World Journal of Microbiology and Biotechnology*, *AMB Express*, *Frontiers in Bioengineering and Biotechnology*, and *Biodegradation*, alongside a book chapter on soil biodiversity and climate change. She has served as Co-PI on national research projects, including an ASRT-NBNE grant on COVID-19 and biotechnology research, and has collaborated on several international projects with institutions in Saudi Arabia.

Recently, she conducts research on Black Soldier Fly Larvae (BSFL)-associated microorganisms and their applications in sustainable agriculture and environmental biotechnology with Prof. Longyu Zheng.

Dr. Halema has authored and co-authored numerous publications in high-impact international journals and has collaborated with researchers from China, Saudi Arabia, and German. She has participated in several nationally and internationally funded research projects and actively contributes as a reviewer for scientific journals. Her long-term research vision is to integrate multi-omics technologies, bioinformatics, and AI-driven approaches to develop innovative solutions for precision agriculture, environmental sustainability, food security, and the circular bioeconomy.

#### **From Contaminated environment to Sustainable Solutions: Microbial Genomics for Heavy Metal Bioremediation and Plant Growth Promotion**

Heavy metal pollution from industrialization, mining, and improper waste disposal threatens ecosystems, agriculture, and human health, making sustainable microbial bioremediation an attractive alternative to conventional remediation methods. This work integrates microbial isolation, physiological characterization, electron microscopy, whole-genome sequencing (WGS), comparative genomics, structural bioinformatics, and gene expression analysis to elucidate heavy metal resistance and plant growth promotion mechanisms in environmentally important bacteria isolated from industrial wastewater and rhizospheric

soil. Among the characterized strains, *Enterobacter roggenkampii* FACU2 removed 62% of cadmium, with significant induction of *czcA*, *cadA*, *czcC*, and *czcD* resistance genes under cadmium stress. *E. kobei* FACU6 achieved 83.4% lead removal (571.9 mg/g biomass) and tolerated up to 3,000 mg/L Pb, with resistance genes upregulated under optimal growth conditions. *E. cloacae* FACU1 (Ars9) removed 85.5% of arsenic (1,415 mg/g biomass), while a second *E. cloacae* strain tolerated up to 15,000 ppm arsenic; structural modeling (SWISS-MODEL, AlphaFold2) and docking confirmed ArsC-mediated arsenate reduction and ArsB/Acr3-mediated arsenite efflux. *Raoultella planticola* FACU3 combined high lead tolerance with plant growth-promoting (PGP) genes for IAA, phosphate solubilization, and phenazine production. Separately, *Escherichia coli* FACU2024 was identified as a dual-function PGPR, harboring 142 growth-promotion genes and 15 heavy-metal resistance genes. SEM/TEM imaging confirmed metal adsorption, intracellular accumulation, and stress-related morphological changes across strains. Genome sequencing (4.8–5.7 Mb) revealed diverse metal-resistance genes, efflux systems, genomic islands, and mobile genetic elements underlying environmental adaptability. Collectively, these findings establish a genomic framework identifying dual-function bacterial strains with strong potential for environmental bioremediation, sustainable agriculture, and next-generation biofertilizer development.



**Dr. Sameh Mohamed Mohamed El-Sawy** is an Associate Professor at the Agricultural and Biological Research Institute, National Research Centre (NRC) in Cairo, Egypt. He holds a Master's degree in Strawberry Cultivation and Productivity and a PhD in Rationalizing Water Consumption for Tomato Plants under Arid/Sandy Soil Conditions (2016). His international academic record features significant participation in China, including three professional training programs and a prestigious tenure under the Talented Young Scientist Program (TYSP) at Ningxia University (2017-2018), where he focused on smart saving-irrigation technology in arid regions and conducted field trials in Tongxin County. A seasoned researcher, Dr. El-Sawy has contributed to 15 local and 6 international projects, serving as the Principal Investigator for many projects; 1- the NRC-funded project on deficit irrigation techniques, 2- using modern technology in greenhouses for producing seedlings and vegetable crops in soilless cultures and 3- sustainable development for ASRT New Valley farm using new technology funded by Academy of Scientific Research and Technology (ASRT). Furthermore, Dr. Sameh worked as Corresponding Investigator for aquaponics systems in saline arid regions (ASRT-funded), and climate-smart practices project funded by

ANSO (China). He also Executive Manager for a multi-year Mediterranean greenhouse optimization initiative international project (PRIMA-funded). Moreover, he is authored 38 papers in peer-reviewed international journals, contributions to 15 national and international conferences, and active supervision of postgraduate students in Egypt and China, he also leads agricultural operations as the Head of Vegetable Production at the NRC Farm in El-Noubaria and Director of the Model Research Farm in Al-Moghra under the Academy of Scientific Research and Technology.

#### **Precision Agriculture and Smart irrigation: Mitigating Water Scarcity and Enhancing the crop productivity under climate Change**

Global agriculture faces an unprecedented dual challenge: meeting the food demands of a rapidly growing global population—projected to reach nearly 10 billion by 2050—while simultaneously mitigating the adverse impacts of climate change, resource depletion, and water scarcity. Traditional farming practices are no longer sufficient to sustain productivity under these volatile environmental dynamics. Egypt faces severe challenges to its food security, primarily driven by its arid climate and low annual rainfall. These natural constraints are further exacerbated by climate change, which negatively impacts crop productivity and plant growth. Consequently, maximizing plant water-use efficiency through advanced irrigation technologies and water-saving methods has become imperative. Agriculture currently accounts for approximately 70% of global freshwater withdrawals. However, traditional irrigation techniques, such as flood or standard overhead sprinkler systems, often result in significant water loss through evaporation, runoff, and deep percolation. To overcome these structural inefficiencies, the agricultural sector is undergoing a digital transformation often referred to as Agriculture 4.0. By leveraging advanced digital tools, sensor networks, and predictive algorithms, smart agriculture shifts farming from a reactive, intuition-based activity to a proactive, data-driven science. Among these technological innovations, precision irrigation stands out as a crucial intervention for climate adaptation and resource conservation. Smart agriculture and precision irrigation are no longer technological luxuries; they are essential frameworks for ensuring global food security under mounting environmental pressures. By combining real-time IoT monitoring, satellite-based remote sensing, and predictive AI, modern agriculture can produce significantly more yield while consuming drastically fewer resources.



**Dr. Sahar Abd EL-Aziz Othman** is a Researcher at the National Water Research Center (NWRC), Central Laboratory for Environmental Quality Monitoring (CLEQM), Egypt. She earned her Ph.D. in the field of wastewater treatment and has over three decades of professional experience in water and environmental research.

She began her career at the Drainage Research Institute (DRI), NWRC, where she worked from 1996 to 2001. Since 2001, she has been working at the Central Laboratory for Environmental Quality Monitoring (CLEQM), NWRC, contributing to research and environmental monitoring programs.

Her expertise includes chemical, physical, and biological analyses of water, soil, and plant samples, with extensive experience in environmental quality assessment and laboratory analysis. Her current research focuses on water and wastewater treatment using innovative chemical and biological materials, as well as water quality assessment through the application of mathematical water quality models.

Her research interests include sustainable water treatment technologies, environmental monitoring, water quality modeling, wastewater reuse, and the development of environmentally friendly treatment approaches to support water resource management

### **Mitigating Severe Salinity Stress in *Vicia faba* L. via Exogenous Proline**

Climate change and its impacts on soil salinization threaten faba bean (*Vicia faba* L.) production in the Nile Delta. Although the osmotic benefits of exogenous osmoprotectants are well recognized, long-term, tissue-specific partitioning mechanisms of repeated foliar proline under unmitigated severe saline irrigation (12 dS m<sup>-1</sup>) are poorly quantified. In the study, a holistic hydrochemical and agronomic approach was used to evaluate the efficacy of weekly foliar proline (500 mg L<sup>-1</sup>) as an intervention for climate-resilience. The hydrochemical modeling confirmed severe irrigation constraints at 12 dS m<sup>-1</sup> indicating strong reverse base-exchange phenomena.. Weekly proline applications reprogrammed the stress response, with pod counts returning to 33.67 and 91.75% reproductive potential preserved. Under moderate salinity (6 dS m<sup>-1</sup>), proline application recovered 70.91% of the yield potential lost due to osmotic stress. Metabolic recovery was governed by a coordinated N-â€œprotein trait axis, as indicated by a multivariate principal component analysis (97.47 % of the variance explained). The study offers a robust data-driven framework to transform hazardous saline water into an agricultural input and a scalable climate-adaptation approach aligned with SDGs 2, 6, and 12.



**Dr. Heba Abdelaziz Moussa AbdAlla** is a Researcher at the Botany Department, National Research Centre (NRC), Egypt, specializing in plant biotechnology, tissue culture, molecular biology, and sustainable agricultural innovation. Her research integrates advanced biotechnology, genomics, and molecular breeding to develop climate-resilient crops capable of withstanding heat, drought, and salinity stresses, contributing to global food security and sustainable agricultural systems.

She has established an active international research profile through strategic collaborations with leading institutions, particularly the Chinese Academy of Sciences (CAS), and has played a key role in developing joint Egypt-China research initiatives and international scientific partnerships. Her work spans plant tissue culture, crop genetic improvement, controlled-environment agriculture, carbon footprint assessment of plant production systems, and the sustainable utilization of plant resources.

Beyond research, Dr. Abdalla is actively engaged in science diplomacy, research management, and the organization of international scientific conferences that foster interdisciplinary collaboration and knowledge exchange. She has successfully contributed to numerous competitive research proposals, international collaborative projects, and scientific publications. Driven by a passion for translating scientific discoveries into practical solutions, she is dedicated to advancing sustainable agriculture, strengthening global research partnerships, and promoting innovation that addresses the pressing challenges of climate change and food security.

**Reproductive-Stage Heat Stress Tolerance in Rice: Implications for Climate-Resilient Agriculture under Global Warming**

Publication Link:

<https://advanced.onlinelibrary.wiley.com/doi/10.1002/advs.202513169>

Climate change represents one of the most critical challenges to global agricultural sustainability, particularly through rising temperatures, increasing drought frequency, and soil salinization, which severely threaten crop productivity and food security. My recent research on rice investigates the physiological, cellular, and molecular mechanisms underlying reproductive-stage heat stress tolerance, a critical determinant of yield stability under warming climates.

Using an integrated approach that combines tissue culture-based screening, controlled environmental stress assays, and molecular analyses, I evaluated genotype-specific responses to elevated temperature, focusing on pollen viability, spikelet fertility, and yield-related traits. The results revealed substantial genetic variation among Egyptian and international rice germplasm, enabling the identification of heat-resilient genotypes with superior reproductive performance under thermal stress. Heat exposure significantly affected pollen development, anther dehiscence, and fertilization efficiency, which were closely associated with differential expression patterns of stress-responsive and chloroplast-associated genes.

These findings provide valuable molecular targets and physiological markers for breeding programs aimed at enhancing heat tolerance in rice. The study further highlights the importance of integrating advanced phenotyping platforms, molecular breeding strategies, and data-driven approaches to accelerate the development of climate-smart crop varieties. Overall, this work contributes to sustainable agricultural strategies by improving crop resilience to climate extremes, enhancing yield stability, and supporting food security in vulnerable regions, particularly in arid and semi-arid environments. Moreover, the research underscores the critical role of international collaboration in advancing climate adaptation strategies through scientific innovation, technology transfer, and knowledge exchange.



**Dr. Mai Nossair Amer** is a Researcher of Chemistry of Natural and Microbial Products at the National Research Centre (NRC), Egypt, with over 15 years of scientific expertise in microbiology, biotechnology, and functional probiotics. Holding a BSc in Biotechnology, an MSc in Prophylactic and Therapeutic Probiotics, and a PhD focused on anticancer bioactive compounds from probiotics, she has built a distinguished research career spanning microbial

genetics, natural bioactive molecules, food microbiology, and sustainable bio-based solutions. Dr. Amer has contributed to numerous national and international projects on probiotics, microbial therapeutics, environmental biodegradation, and biological plant protection, and has authored multiple indexed scientific publications in high-impact journals. Alongside her research, she teaches food microbiology and infection diseases at Helwan University and is an active member of OWSD and several scientific networks. As an emerging entrepreneur CEO and founder of AgriFood Serv, she bridges science with innovation by transforming microbial research into sustainable agricultural and food-industry applications.

### **From Lab Innovation to Agricultural Entrepreneurship: Lactic Acid Bacteria for Sustainable Farming and Healthy Food Systems**

Transforming scientific research into real-world agricultural and food solutions is essential for building sustainable and healthy communities. This presentation explores how lactic acid bacteria (LAB) can move beyond laboratory research to become a driving force in agricultural entrepreneurship and functional food innovation.

Lactic acid bacteria are well known for their applications in food fermentation and probiotic products. However, recent advances demonstrate their powerful role in agriculture as biofertilizers, biostimulants, and biological control agents. Through the production of organic acids, antimicrobial metabolites, enzymes, and plant-growth-promoting compounds, LAB improve soil fertility, enhance nutrient availability, suppress plant pathogens, and increase crop resilience to environmental stress.

By integrating microbiological research, strain development, and scalable production systems, LAB-based technologies can support sustainable farming practices while simultaneously contributing to healthier food chains with reduced chemical residues. This approach connects agricultural productivity with food safety, environmental protection, and market competitiveness.

The presentation highlights how science-driven entrepreneurship in agricultural biotechnology can empower women scientists to create innovative enterprises that strengthen food security, promote healthy nutrition, and advance sustainable development.





**Dr. Maissara Mohamed Kamal Elmaghraby** is a dedicated Researcher at the Central Laboratory of Organic Agriculture, Agricultural Research Center (ARC), Giza, Egypt. With over 15 years of professional experience in organic and agricultural microbiology, he has contributed significantly to advancing sustainable agricultural practices in Egypt.

He earned his B.Sc. in Agricultural Cooperative Sciences (Agribusiness and Marketing Department) from the Higher Institute for Agriculture Cooperation (2006). He completed his Pre-M.Sc. studies in Agricultural Microbiology at Cairo University (2009, Excellent), followed by a Master's degree (2015) focusing on "PGPR as Enhancer of Mint under Organic Agriculture System." He later obtained his Ph.D. in Agricultural Microbiology (2021) from Benha University, with research on organic farming applications to increase vegetable crop productivity.

Since 2008, he has worked extensively in soil and plant microbiology, compost and vermicompost production, greenhouse and field applications, biofertilizers, and biological control of nematodes and soil-borne fungi.

In addition to research, Dr. Elmaghraby is actively involved in academic training and community outreach. He supervises practical training for agriculture students, provides technical consultations for graduation projects, trains farmers and organic companies, and participates in national initiatives promoting organic agriculture, biofertilizers, and composting technologies across Egyptian governorates."

### **Lactic Acid Bacteria-Based Bio-Stimulant Production from Low-Quality Date Waste: A Sustainable Circular Bioeconomy Approach**

The accumulation of low-quality date palm ( *Phoenix dactylifera* L.) fruits represents a

significant environmental and economic challenge in major date-producing regions, including Egypt and the Middle East. Large quantities of downgraded or postharvest-damaged dates are often discarded, contributing to organic waste buildup, greenhouse gas emissions, and loss of valuable bioresources. This study proposes an innovative biotechnological approach utilizing lactic acid bacteria (LAB) as a bio-stimulant product to recycle low-quality date waste into value-added agricultural inputs. Selected LAB strains, were employed to ferment date waste under controlled conditions. The fermentation process enhanced the release of bioactive compounds such as organic acids, amino acids, phenolic compounds, and soluble nutrients, producing a biologically enriched formulation suitable for plant application.

The resulting LAB-based bio-stimulant demonstrated potential to improve nutrient availability, stimulate root development, enhance microbial soil activity, and increase plant tolerance to abiotic stress. Recycling date waste through LAB fermentation not only reduces environmental pollution and landfill burden but also transforms agro-industrial by-products into sustainable agricultural solutions. Economically, this approach supports circular bioeconomy principles by creating new revenue streams from otherwise discarded materials, lowering production costs for bio-stimulant manufacturing, and increasing farm productivity.

Overall, the integration of LAB biotechnology with date waste valorization offers an eco-friendly, cost-effective strategy that aligns with sustainable agriculture goals, promotes waste reuse, mitigates environmental impact, and enhances agricultural resilience in arid and semi-arid regions.

### **Session 3 Environment & Agriculture: Wednesday, 9<sup>th</sup>, 11am-12.30am (1.5h) National Research Center.**



**Prof. Dr. Salwa K. Hassan** is a professor of Air Pollution Chemistry and the head of Air Pollution Research Department, Environment and Climate Change Research Institute, National Research Centre, Dokki, Giza, Egypt. She got B.Sc. degree in 1991 from Department of chemistry, Faculty of Science, Cairo University. She joined the National Research Centre, Air pollution Research Department, Environment Research Division, as a Researcher Assistant in 1994. In 2000, she promoted to Assistant Researcher after obtaining her M.Sc. from Department of chemistry, Faculty of Science, Cairo University in the field of Air Pollution Chemistry. She was promoted to Researcher position in 2007 from Department of chemistry, Faculty of Science, Cairo University. In 2013, she has been promoted to Associate

Professor in the field of Air Pollution Chemistry. In 2019 she was promoted to a full Professor in the same field. On 2024 she becomes the head of air pollution research department. Her main field of interest is air pollution monitoring and measurements; atmospheric photochemical reactions; study the chemistry of the atmosphere and its chemical composition, formation of fine aerosol and its physical characterization and impact of pollution and environmental exposures on health. She has about 34 years of experience in the atmospheric chemistry field. She has organized workshops and participate in national and international conferences. She has published 51 scientific articles cited in SCI journals. She has co-edited a series of books on air pollution, human health and the environment. She has participated in national and international projects as PI and as member. She also reviewer to many international journals and is member of several international scientific committees. Finally, she received the Academy of Scientific Research Award in the field of Environmental Research and Environmental Education.

### **Assessing Indoor Pollution: Organic and Carcinogenic compounds; Heavy Metal Exposure, and Health Risks in Indoor Environments**

This presentation highlights the pivotal role of women researchers in environmental health by examining indoor pollution and its health implications for families. Drawing from empirical studies on domestic and indoor spaces, this research evaluates heavy metals and organic pollutants to which women and children are daily exposed. The first study in primary schools revealed that urban environments exhibit elevated levels of polycyclic aromatic hydrocarbons (PAHs), driven primarily by traffic emissions. While ingestion remains the primary exposure pathway for children, cumulative cancer risks remained within acceptable baselines. The second study in residential houses assessed indoor dust loading rates and heavy metal concentrations (lead, cadmium, and nickel). Indoor dust loading averaged 1.58 g/m<sup>2</sup>/week, with unpaved side streets significantly increasing dust accumulation, whereas main-street proximity amplified toxic metal loads. Building age and paint degradation further escalated lead concentrations. Notably, environmental tobacco smoke severely degraded indoor air quality, increasing dust loading by 1.7-fold and heavy metal levels up to 2.7-fold, causing over 77% of smoking households to exceed US EPA lead safety guidelines.

These findings confirm that domestic spaces pose hidden environmental health risks. Through this field research, practical public health strategies (such as regular wet cleaning, banning smoking indoors, and using lead-free materials) can be developed, enabling scientists to make crucial progress in protecting domestic environments and community health.



**Dr. Alyaa Ibrahim Salim** is a full-time assistant professor at the Faculty of Engineering and Applied Science, Nile University (NU). She received the B.Sc. (2006) in chemistry, faculty of science, Mansoura University, Egypt. She got a diploma in biochemistry, Alexandria University (2007-2008) and diploma in Analytical Chemistry, Cairo University (2010). She received the M.Sc. (2015) and Ph.D. (2018) degrees in analytical chemistry, chemistry department, faculty of science, Cairo University, Egypt. She has publications distributed between high-impact journals, conferences, and book chapters. Her research tracks cover health care, climate changes and solutions, different applications of nanotechnologies, food spoilage, and environmental science. Her research interests include pharmaceutical analysis, electrochemical analysis, spectroscopic techniques, HPLC techniques, quantum analysis, and sensors fabrication. She is a member in the Egyptian Society of Analytical Chemistry, Cairo university, Egypt.

#### **Synergistic CO<sub>2</sub> Capture: A Novel Chitosan-Metal Oxide Nanoparticle-MOF Hydrogel Composite from Waste-Derived Materials**

This study presents a novel composite hydrogel bead system for efficient CO<sub>2</sub> capture, synthesized entirely from green, waste-derived materials. Chitosan serves as the biodegradable matrix, integrated with metal oxide nanoparticles (MO NPs) from organic waste and metal-organic frameworks (MOF) as active nano-adsorbents. Beads were fabricated via dropwise coagulation. Characterization confirmed successful integration: XRD verified crystallinity, FTIR showed chemical bonding, and SEM revealed a transformation from smooth chitosan to highly porous, granular composites, indicating increased surface area. EDX quantitatively confirmed nanomaterial incorporation.

CO<sub>2</sub> capture performance, evaluated using a custom laboratory system with colorimetric monitoring, demonstrated exceptional results. The composite beads achieved 100% capture efficiency for 50 mL CO<sub>2</sub>, outperforming chitosan-only (15%), MO NP-only (55%), and MOF-only (75%) beads. Post-capture analysis provided direct evidence of chemisorption: FTIR showed carbonate species formation, SEM indicated surface densification, and EDX confirmed a carbon content increase from 14.66 wt% to 27.81 wt%.

The integration of waste-derived MO NPs and MOF into chitosan creates a synergistic, high-capacity, reusable sorbent. This system combines MO NP chemisorption, MOF physisorption, and chitosan's structural versatility. These green-synthesized hydrogel beads offer a scalable, sustainable approach for industrial point-source CO<sub>2</sub> capture, supporting circular economy principles and decarbonization pathways.



**Dr. Shymaa Sabry Mohamed Zaher**, Assistant Professor of Phycology, Fresh water and lakes division, Inland water branch, National Institute of Oceanography and Fisheries (NIOF), Egypt. her research interests concern the Botany specially phycology. Algal biotechnology, Usage of applied algae in the medical field, cosmetics and different scientific fields. Special expert in Algal Identification in marine environment. Special concern on toxic algal blooms. Membership in The Egyptian phycological society, Egyptian net for climate change CAN EGYPT 2030. Participating in over than 15 research projects at various aquatic environments.

**Assessing the Therapeutic efficiency of Spirulina platensis on ethanol-Induced Oxidative Markers, DNA Damage and some Organs histopathology in Rat**

The therapeutic antioxidant properties of Spirulina platensis conquered scientific interest. The present study assesses the possible antioxidant activity and DNA fragmentation of Spirulina platensis powder against ethanol-induced gastric ulcers in rats. To explore the possible efficiency, antioxidant defense enzymes (e.g., superoxide dismutase, and glutathione peroxidase), were examined as well as the level of lipid peroxidation (MDA) and the DNA damage in the gastric mucosa by comet Assay. Intragastric administration of Spirulina (500 mg/kg body weight) before (protected) or after (treated) injection with ethanol significantly lowered the ulcer index value compared to the ulcer control group ( $p < 0.05$ ). Moreover, the alcohol-induced decrease in stomach antioxidant enzyme activity found in the ulcer control group was prevented by Spirulina powder post-treatment or pretreatment. Furthermore, it reversed the ethanol-induced increase in lipid peroxidation and DNA fragmentation. Also, the focus was on assessing the impact of these compounds on histopathological lesions induced by ethanol in the liver, kidney, and lung of rats. On conclusion, Spirulina recorded, anti-ulcerative, anti-inflammatory, antioxidant, and diminish DNA damage. Its healing action may be used as alternative treatment gastric ulcer instead using chemical drugs and has similar effect as organic tablets.



**Dr. Douaa Hussein Abdel Aziz** started working at the National Research Center by 2008. She had her MSc in Microbiology in 2014 from Ain Shams University and her PhD in Microbiology from Ain Shams University in 2019. She focused on working in microbial ecology, bioremediation for pollution reduction, and microbial treatment of environmental hazards to reduce their harmful effect on the environment and humans as well. She published five



papers; three of them are indexed in Scopus. She was awarded the Best Master Thesis—Agricultural Professions Syndicate, 2015. She was a speaker at the International Conference of the Faculty of Science, Ain Shams University, "Scientific Innovations and Sustainable Development (SISD)" – 2018. She has been a reviewer in the International Journal of Microbiology and Biotechnology (IJMB) since 2021. She is a member of the American Society of Microbiology and the Society of Applied Microbiology, 2025.

### **Bioremediation and Our Lives: Restoring the Environment We Depend On**

Reducing contamination in our environment is a crucial step to improve our health and well-being. Coal is considered one of the world's primary energy sources and a priceless natural resource. It has played a key role in economic growth by serving as a major source of energy. Although coal mining operations have contributed to economic growth in many countries, it is one of the most dangerous activities in the world, due to high environmental and safety risks. These risks affect soils, plant communities, wildlife, and water sources and release significant amounts of CO<sub>2</sub> and CH<sub>4</sub> into the atmosphere. This contributes to increased greenhouse gases, global warming, and environmental damage that affect human life. In addition, coal mining leads to adverse health effects, including a lot of diseases like ischaemic heart disease, cerebrovascular disease, chronic obstructive pulmonary disease, lower respiratory infections, and lung cancer. The white-rot fungi have garnered significant interest due to their remarkable ability to degrade lignocellulosic materials and decompose some environmental contaminants, such as charcoal. This study aims to evaluate the biodegradation of the fungal strain *Phanerochaete chrysosporium* for the pollutant charcoal and its extreme tolerance to cope with such a pollutant and use it as a sole carbon source to reduce its bad impact on the environment.



**Elham Ahmed Abdelrahim Mohamed** is an archaeological conservator and academic researcher specializing in the conservation and restoration of antiquities, with a focus on mural paintings. She is a recipient of the prestigious "Scientists for Next Generation" (SNG) grant from the Academy of Scientific Research and Technology (ASRT) in Egypt. Elham graduated with a Bachelor's degree (B.A.) in Archaeology with a grade of "Excellent with Honors" from Fayoum University. She subsequently obtained her Master's degree (M.Sc.) in Conservation from the same institution with a grade of "Excellent with a formal recommendation for publication." Her research interests lie at the intersection of cultural heritage preservation, nanotechnology, and advanced computer applications,

particularly the integration of Artificial Intelligence and Machine Learning in preventive conservation and climate-adaptive heritage management.

### **The Role of Artificial Intelligence in the Preservation of Mural Paintings Under Climate Change: Opportunities, Challenges and Emerging Frameworks**

This study provides a structured analytical framework to elucidate how AI-driven methodologies can fundamentally shift preventive conservation strategies from a paradigm of reactive repair to a proactive, sustainable stewardship approach under climatic stress. By conducting a systematic examination of the integration of Machine Learning (ML), Deep Learning (DL) and the Internet of Things (IoT), the research paper critically evaluates the efficacy of these technologies within real-world heritage scenarios. The analytical results highlight the transformative capability of Convolutional Neural Networks (CNNs) for identifying microcracks and surface defects within controlled environments with remarkable precision, achieving accuracy rates of 94%. Concurrently, it explores the potential of digital twins in simulating non-invasive restoration interventions, thereby minimizing physical risks to the artwork. findings demonstrate that adopting AI-driven predictive maintenance strategies offers significant economic benefits, potentially reducing long-term conservation costs by approximately 35% through the implementation of early warning systems. Consequently, this paper establishes a strategic roadmap for integrating AI into heritage management, emphasizing the need to adopt "Green AI" principles and establish ethical governance frameworks. It concludes that fostering interdisciplinary collaboration is essential for building resilient, climate-adaptive systems capable of safeguarding cultural heritage for future generations.



**Dr. Arig Wahid Elkhoully** is a researcher and lecturer in Food Processing Technology at Borg El Arab Technological University (BATU), Egypt. She holds a Ph.D. in Environmental Studies, a Master's in Dairy Science and Technology, and a Bachelor's in Food Science and Technology from Alexandria University.

Her research focuses on sustainable food systems, valorization of agro-industrial wastes, and upcycling processing by-products into high-value functional ingredients within circular economy frameworks. Dr. Elkhoully's work is published in leading international journals, including Nature Scientific Reports and Springer's Discover Food, reflecting her dedication to zero-waste food technologies and sustainable nutrition.



Dr. Elkhoully is a Certified ISO 9001 Lead Auditor with eight years of industrial experience as a production engineer in the packaging sector (2012–2020). She has also served as an Exam Invigilator for the British Council in Alexandria since 2012. Recognized as an ambassador for climate action, water conservation, risk management, and governance, she actively participates in international scientific forums and environmental initiatives, bridging academic research, industrial application, and sustainable development goals.

### **Hazelnut Shell Powder as a Sustainable Novel Functional Ingredient for Quality Improvement of Crackers**

To address the challenges of agro-industrial wastes, agri-food by-products represent valuable, underutilized sources of bioactive compounds for sustainable processing. This study investigated the valorization of hazelnut shells a primary agro-industrial waste by incorporating hazelnut shell powder (HSP) into baked crackers through partial wheat flour substitution at levels of 0%, 2.5%, 5%, 7.5%, and 10% (w/w). Physicochemical evaluations revealed that HSP fortification significantly ( $P < 0.05$ ) enhanced crude protein, ash, crude fiber, cellulose, hemicellulose, lignin, and essential minerals (Ca, Zn, Mn, Cu). Furthermore, HSP integration markedly increased total phenolic and flavonoid contents, leading to elevated DPPH and ABTS radical scavenging antioxidant activities. Texture profile analysis demonstrated an increase in hardness, gumminess, cohesiveness, and chewiness alongside higher substitution levels. Sensory evaluation confirmed that moderate incorporation levels preserved overall consumer acceptability while delivering a nutrient-dense, fiber-enriched functional snack. These findings demonstrate the strong potential of upcycling agro-industrial processing waste into functional food ingredients, offering an effective strategy for waste reduction and the development of value-added healthier food products. This research directly supports sustainable food systems and aligns with Sustainable Development Goal 12 (Responsible Consumption and Production). Keywords: Agro-industrial waste, Upcycling, Hazelnut shells, Functional food, Crackers, Circular economy, SDG 12 also can select SDG 9: Industry, Innovation, and Infrastructure for sustainable food innovation).